

Estimation Report

Intermediate COCOMO for 44 KLOC, semi-detached, EAF = 1.15

Project	Course	Case Study	Version
HaulTrack - 1,800-truck fleet telematics	B.Tech CSE - Software Engineering & Project Management	No. 19	1.0 (baseline)

1. Inputs

Item	Value	Source
Size	44 KLOC	Given
Mode	Semi-detached	Given: mixed team and experience, some tight and some flexible constraints
EAF	1.15	Given (cost-driver ratings below multiply to 1.152, which rounds to 1.15)

2. Model constants (Boehm, COCOMO 81)

Model	Effort (person-months)	Duration (months)	Coefficients, semi-detached
Basic	$E = a (KLOC)^b$	$D = c (E)^d$	$a = 3.0, b = 1.12, c = 2.5, d = 0.35$
Intermediate	$E = a (KLOC)^b \times EAF$	$D = c (E)^d$	$a = 3.0, b = 1.12$ (Boehm's intermediate table), $c = 2.5, d = 0.35$

3. Cost drivers and justified ratings

Code	Driver	Rating	Multiplier	Justification
RELY	Required reliability	High	1.15	A wrong score can cost a driver income and a wrong alert wastes dispatch effort: high financial loss, but no direct loss of life, so High not Very High.
DATA	Database size	Very High	1.16	Test database / program size: even one day of replay data (about 1 GB, see Design Pack) divided by 44 KLOC is far above 1,000, the Very High threshold.
CPLX	Product complexity	High	1.15	Stream processing, map-matching, terrain normalisation and statistical scoring; more than routine CRUD, short of real-time control.
TIME	Execution time constraint	High	1.11	Ingest must keep pace with about 60 records/s average (higher at peaks); target use of about 85% of the time budget at peak.
STOR	Main storage constraint	Nominal	1.00	Elastic cloud storage; no tight memory limits.
VIRT	Virtual machine volatility	Nominal	1.00	Stable runtime; major platform change roughly every 6-12 months.
TURN	Computer turnaround time	Nominal	1.00	CI builds and test runs in minutes to hours, typical.
ACAP	Analyst capability	High	0.86	Requirements are demanding; assume 75th-percentile analysts are assigned to the conflict and fairness work.
AEXP	Applications experience	Nominal	1.00	Team has about 3 years of average telematics / logistics exposure.
PCAP	Programmer capability	Nominal	1.00	Mixed team, median capability.
VEXP	Virtual machine experience	Nominal	1.00	About 1 year on the chosen platform.
LEXP	Language experience	High	0.95	Team is fluent in the implementation languages (about 3+ years).
MODP	Modern programming practices	High	0.91	Code review, CI, automated tests are mandatory (the regression gate depends on it).
TOOL	Use of software tools	High	0.91	Integrated CI, static analysis and test-data tooling are provided.
SCED	Required schedule	Nominal	1.00	Schedule follows the model's nominal; no compression.

Code	Driver	Rating	Multiplier	Justification
	Product (EAF)		1.152	Rounds to the given 1.15, which is the value used below.

4. Calculation

$KLOC^b = 44^{1.12} = 69.29$
 Nominal E = $3.0 \times 69.29 = 207.9$ PM
 Intermediate E = $207.9 \times 1.15 = 239.0$ PM
 Intermediate D = $2.5 \times 239.0^{0.35} = 17.00$ months
 Average team = $E / D = 239.0 / 17.00 = 14.1$ FTE

5. Comparison with the basic model

	Basic	Intermediate	Difference
Effort (PM)	207.9	239.0	+31.2 (+15.0%)
Duration (months)	16.19	17.00	+0.81 (+5.0%)
Average staff (FTE)	12.8	14.1	+1.2

Why they differ. For semi-detached software the basic coefficients (3.0, 1.12) are identical to the intermediate nominal coefficients, so the whole gap comes from the EAF. The basic model silently assumes every driver is Nominal (EAF = 1.00). HaulTrack is not nominal: reliability, complexity, execution-time and especially database size all push cost **up**, and only capable analysts, modern practices and tools pull it back **down**. The net 1.15 adds 31.2 person-months. Duration moves less than effort (+5.0% versus +15%) because duration grows only with $E^{0.35}$, so extra effort is absorbed mostly by a larger team, not a longer schedule.

Sanity check and use. An average of about 14 FTE over roughly 17 months (about 74 weeks) is plausible for 44 KLOC. The Project Plan staffs in phases (smaller during requirements and design, peaking at about 20 during build and test) to the same 239 PM total, and its Gantt chart spans 74 weeks to match. COCOMO assumes a nominal schedule; compressing it would raise cost (SCED rating) and is not planned. The estimate carries the usual +/-20-30% uncertainty; the biggest sensitivities are size (44 KLOC) and the DATA and TIME ratings. It will be re-baselined at the end of design (milestone M2).

WBS allocation. Effort is distributed across the ten WBS packages in the Project Plan so the allocations sum to the estimated total (rounding: 239.0 vs 239.1 PM).